

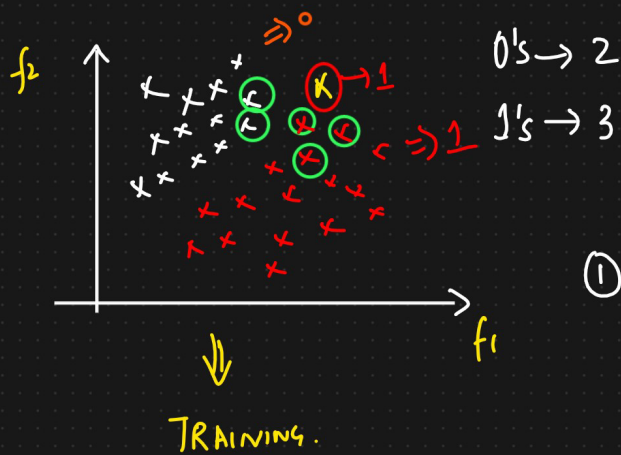
K Nearest Neighbour (KNN)

① Classification

② Regression

① Classification

K=5



f_1 f_2 y [Binary categories]

- - 0
1

① We have to initialize the K value

K=5

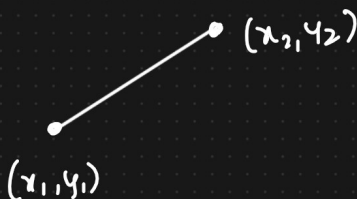
$K > 0 \dots \infty$

$K=1, 2, 3, 4, 5 \dots \Rightarrow$ hyperparameter

② Find the K Nearest Neighbour for
The Test Data.

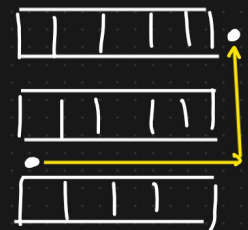
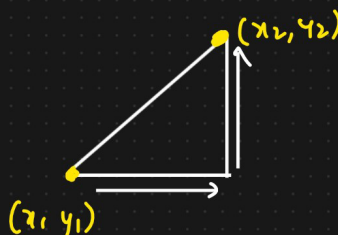
③ From those $K=5$ how many
neighbour belong to 0 category
and 1 category

① Euclidean Distance



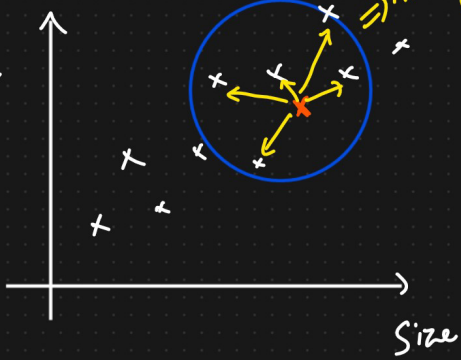
$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

② Manhattan Distance



② Regression

House
price



Median
 $\Rightarrow$ Average of all these
points to find out the
o/p,



Time complexity

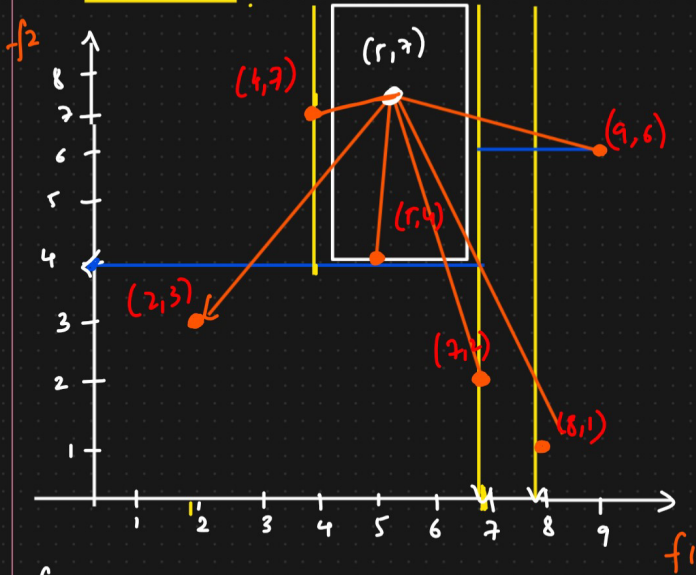
$O(n) \Rightarrow$ Million of data point

{ ① KD Tree
② Ball Tree } $\Rightarrow$ Optimize

$\Downarrow$
Binary Tree

Variants of KNN

KD Tree



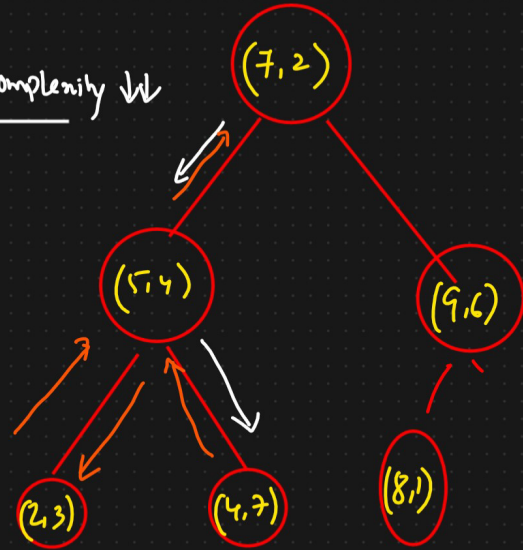
f_1
 2, 4, 5, 7 | 8, 9
 $\frac{5+7}{2} = \frac{12}{2} = 6.5$

f_2
 1, 2, 3, 4 | 6, 7

f_1	f_2
7	2
5	4
9	6
2	3
4	7
8	1
Median	Median

Binary Tree [KD Tree]

Time Complexity ↓



Back Tracking

(2) Ball Tree

Time Complexity ↓

